

DS Report: Trader Sentiment & Performance Analysis

1. Introduction

This project investigates the relationship between crypto market sentiment (Fear & Greed Index) and trader performance. The objective is to uncover patterns and actionable insights by analyzing both historical sentiment data and trader transactions.

2. Data Preparation

- Downloaded and merged market sentiment and trader data using trade date as key.
- Cleaned missing values and parsed relevant columns to correct data types.
- Created additional features including sentiment classification and profit/loss indicators.

3. Key Findings

- **Market Sentiment Distribution:** Most trading activity took place during "Fear" and "Extreme Fear," reflecting longer or more frequent occurrence of these market periods.
- **Weak Linear Correlation:** There is a very weak linear correlation (0.01) between the index value and closed PnL, meaning the raw index value does not reliably predict trade outcome.
- **Profitability by Sentiment:** Trades closed during "Extreme Greed" achieved the highest average closed PnL. "Neutral" periods had the lowest average trade profitability.
- **Trading Activity vs. Profitability:** Highest trading volume occurred during "Fear" sentiment, yet "Extreme Greed" periods gave better average returns. This suggests volume alone doesn't guarantee profitability—market sentiment matters.
- **Trading Side Distribution:** During "Fear," buy/sell volume was balanced; during "Extreme Greed," there was a slight shift towards sells, but not significant.
- **Coin Performance by Sentiment:** Some coins showed stronger average profitability in "Extreme Greed" periods. Identifying coins that are resilient or thrive in bullish sentiment could be strategically useful.

4. Strategy Insights

- Focus on sentiment classification, not just index values, to inform trades.
- Consider counter-sentiment approaches (caution during high-volume fear, and capitalize during greed).
- Tailor coin selection to sentiment, using historical performance as guidance.

5. Conclusion and Next Steps

The analysis highlights that trading behavior and profitability are sentiment-dependent. While general fear drives volume, greed drives profitability. This analysis suggests trading in line with sentiment classification can improve average returns. Future work could compare trade durations, deep-dive into strategy types used in "Extreme Greed," and broaden risk metrics for further insight.